CST557 - Quantum Computing

CST 557 (Quantum Computing) Syllabus

Course Information

Course Details

- **Course Title:** Quantum Computing
- **Course Number:** CST 557
- **Meeting Location:** Physical
- **Units:** 2

Course Description

This course introduces concepts of quantum computing through theoretical exploration and hands-on experimentation. Students will learn about qubits, superposition, and entanglement, and will implement quantum circuits and algorithms using open-source software tools for quantum simulation. Key topics include quantum gates, entanglement, Shor's and Grover's algorithms, and quantum error correction. By the end of the course, students will have a practical understanding of quantum computing principles and the ability to experiment with quantum circuits.

Materials

- Course textbook: "Quantum Computing: An Applied Approach" by Jack D. Hidary
- Supplemental materials: IBM Qiskit Textbook (online), Microsoft Quantum Development Kit documentation, selected research papers from quantum computing journals, and online courses from quantum computing platforms

Technology Requirements

Students must have access to: - A computer (not "chromebook" or similar) with reliable internet connection - Access to Canvas learning management system - Quantum computing simulation software (Qiskit, Cirq, or similar quantum development frameworks) - Python programming environment with quantum computing libraries - Access to cloud-based quantum computing platforms (IBM Quantum, Google Quantum AI, or similar)

Course Objectives

At the end of this class, you should be able to:

1. Understand fundamental quantum computing concepts including qubits, superposition, and entanglement
2. Design and implement quantum circuits using quantum gates and simulation tools
3. Analyze and implement key quantum algorithms including Shor's and Grover's algorithms
4. Apply quantum error correction techniques and evaluate quantum computing applications

Grading

There are three groups of assignments, briefly described in the table below. **Getting below a 40% in any of these three groups will result in failing the class.**

Assignment Kind	Value
Projects	35%
Exams	35%
Participation	30%

Projects

Projects will take the form of basic quantum circuit design exercises, implementation of one fundamental quantum algorithm (such as Grover's algorithm), and hands-on experimentation with quantum simulation tools. Students will gain practical experience with quantum gates and simple quantum programs using cloud-based quantum platforms.

Exams

Exams will cover material from class and from any assigned reading, as well as draw from knowledge gained during projects.

Participation

Class is expected to be highly interactive, with students actively answering and asking questions, as well as participating in discussion groups. Therefore, attendance is required, as well as participation in all in-class activities. These activities may take the form of in-class discussions, brainstorming sessions, stand-ups, as well as making active progress on projects during class time.

Grade Assignment

Grades as assigned based on the standard ranges, which are outlined below.

Grade Range	Letter Grade
$[97, \infty)$	A+
$[93, 97)$	A
$[90, 93)$	A-
$[87, 90)$	B+
$[83, 87)$	B
$[80, 83)$	B-
$[77, 80)$	C+
$[73, 77)$	C
$[70, 73)$	C-
$[60, 70)$	D
$[0, 60)$	F

Syllabus Change Policy

This syllabus is subject to change at the instructor's discretion to enhance learning or accommodate unforeseen circumstances. Students will be notified of any substantive changes (such as changes to due dates, point values of assignments, or major policy modifications) via Canvas announcement and/or email at least one week in advance when possible.

The process for syllabus changes: 1. Instructor identifies need for change 2. Revised syllabus section is prepared 3. Students are notified via Canvas and email 4. Change takes effect after notification period

Students are responsible for staying current with any announced changes to the syllabus.